

Multiple Choice

1. **Answer:** B

Options: A) three times the bend.; B) twice the bend.; C) four times the bend if the branch doesn't break.; D) no bend at all.; E) a bend that is unpredictable.

Note: Correct option: B - twice the bend.

2. **Answer:** A

Options: A) isotopes; B) ions; C) mesons; D) protons; E) photons

Note: Correct option: A - isotopes

3. **Answer:** A

Options: A) 0.83 diopters; B) 3.3 diopters; C) 5.0 diopters; D) 1.0 diopters; E) 1.2 diopters

Note: Correct option: A - 0.83 diopters

4. **Answer:** C

Options: A) 18.0 s; B) 15.5 s; C) 9.0 s; D) 8.0 s; E) 10.5 s

Note: Correct option: C - 9.0 s

5. **Answer:** A

Options: A) $E^2/4Mc^2$; B) $2E_\gamma^2/Mc^2$; C) E_γ/Mc^2 ; D) $3E_\gamma^2/2Mc^2$; E) $E_\gamma/4Mc^2$

Note: Correct option: A - $E^2/4Mc^2$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '9.05'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'lower'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 25%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 510 cm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key